

Customer Behavior Analysis Report

Prepared for Aflido Tech | Data Analyst Internship Task

Prepared by: Hepsiba Selvi M

Dataset: E-Commerce Customer Behavior Data (250,000 transactions, 49,673 unique customers)

Tools Used: Python, Pandas, Matplotlib, Seaborn (Google Colab)

1. Objective

Analyze customer transaction and behavioral data to identify meaningful customer segments, understand purchase patterns and retention trends, and surface churn risk indicators. The goal is to translate these findings into actionable recommendations that help Aflido Tech improve customer engagement and retention.

2. Methodology

The raw transaction data (250,000 rows) was cleaned and aggregated to the customer level using **RFM analysis** — a standard customer segmentation technique based on three dimensions:

- **Recency** — days since the customer's most recent purchase
- **Frequency** — total number of purchases made
- **Monetary** — total amount spent

Each customer was scored 1–4 on each RFM dimension using quartiles, then grouped into six descriptive segments (Champions, Loyal Customers, Potential Loyalist, At Risk, Needs Attention, and Lost) based on their combined score.

3. Data Cleaning Summary

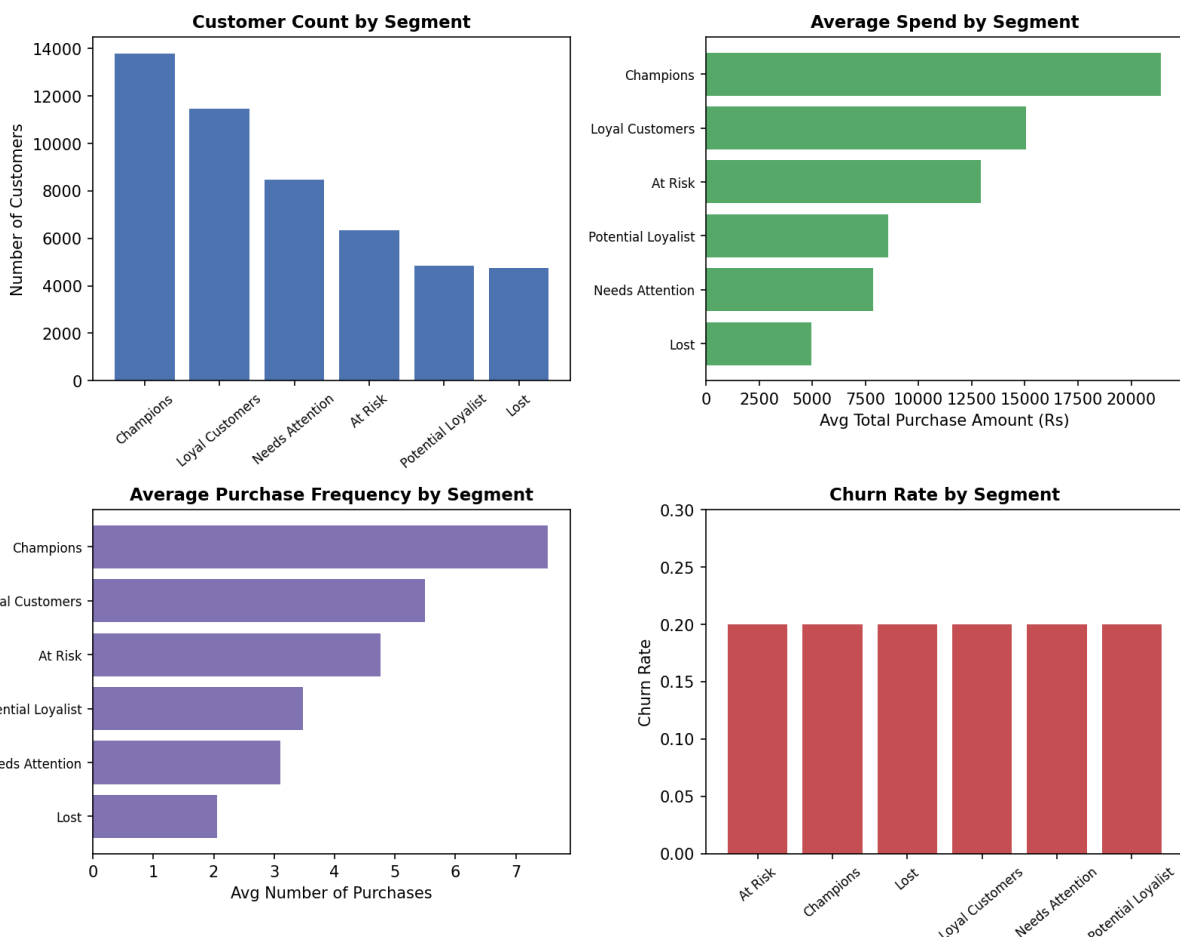
- Converted Purchase Date to proper datetime format
- Filled missing values in the Returns column with 0 (no return recorded)
- Removed a duplicate age column
- Verified zero duplicate rows and zero remaining null values

4. Customer Segments Overview

The 49,673 customers were distributed across six segments as follows:

Segment	Customers	Avg Recency (days)	Avg Frequency (orders)	Avg Monetary (Rs)	Churn Rate	Avg Returns
Champions	13,796	100.53	7.53	21,385.43	20%	3.06
Loyal Customers	11,469	185.94	5.50	15,042.13	20%	2.21
Needs Attention	8,462	378.38	3.10	7,870.47	20%	1.24
At Risk	6,339	382.07	4.76	12,943.87	20%	1.91
Potential Loyalist	4,856	89.71	3.48	8,576.02	20%	1.40
Lost	4,751	717.09	2.05	4,945.25	20%	0.80

5. Visualizations



Top-left: Champions and Loyal Customers together make up almost half the customer base. *Top-right:* Champions spend nearly 4.3x more than Lost customers on average. *Bottom-left:* Purchase frequency is highest among Champions and lowest among Lost customers. *Bottom-right:* Churn rate is flat at 20% across every segment.

6. Key Findings

- Champions (27.8% of customers) generate the highest average spend (Rs 21,385) and purchase most frequently (7.53 orders), making them the most valuable segment to retain.
- Lost customers (9.6%) haven't purchased in an average of 717 days and have the lowest spend and frequency — largely inactive.
- Churn rate is uniform at 20% across all six segments. This indicates that, in this dataset, churn is not explained by RFM behavior alone — a Champion is statistically just as likely to churn as a Lost customer.
- Champions also have the highest average number of returns (3.06), which slightly offsets the margin benefit of their high spend.
- Potential Loyalists are recently active (avg. 89.71 days) but have not yet reached high spend levels, making them a strong target for upsell campaigns.
- Monthly active customers and revenue have remained stable over the multi-year period analyzed, with no major seasonal decline (the final month's dip reflects incomplete data, not an actual drop).

7. Actionable Recommendations for Aflido Tech

1. Reward Champions to protect high-value revenue.

Champions are ~28% of customers but drive the highest average spend. Introduce a loyalty/VIP tier (early access, exclusive discounts) to reduce the risk of losing this group, since their departure would disproportionately hurt revenue.

2. Win back At Risk customers with targeted offers.

6,339 customers haven't purchased in over a year despite a healthy historical spend (avg. Rs 12,943). Personalized win-back emails or discount codes before they fully lapse can recover meaningful revenue.

3. Nurture Potential Loyalists into higher-value segments.

4,856 customers are recently active but haven't spent much yet. Cross-sell and bundle offers based on their past purchase category can convert them into Loyal Customers or Champions.

4. Investigate churn drivers beyond purchase behavior.

Since churn is flat at 20% across every segment, RFM data alone does not explain who churns. Aflido Tech should examine other factors — customer service interactions, delivery experience, or product quality complaints — to identify the real drivers of churn.

5. Reduce returns among high-spend customers.

Champions have the highest return rate (3.06 avg) alongside the highest spend. Reviewing product quality, sizing accuracy, or listing clarity for categories favored by this segment could protect margin without discouraging their purchase volume.

8. Conclusion

RFM segmentation reveals a healthy, well-distributed customer base for Aflido Tech, with Champions and Loyal Customers together contributing the bulk of high-value transactions. The most important finding is that churn is evenly spread across segments, meaning retention strategy should not rely on spend/frequency data alone — deeper investigation into service quality and customer experience is

recommended alongside the segment-specific engagement strategies above.